



2.1 Functions, domain and range

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A function f maps each element of its domain to exactly one element of its range (vertical line test on a graph). The domain is the set of valid input values (excluding, e.g., values making a denominator zero or a square root negative), and the range is the set of resulting output values. Notation $f:x \mapsto f(x)$ or $y=f(x)$ describes the mapping.

Worked example: follow the reasoning

Find the domain and range of $f(x) = \sqrt{x-3}$.
Domain: need $x-3 \geq 0$, so $x \geq 3$.
Range: since $\sqrt{x-3} \geq 0$ for all valid x , range is $f(x) \geq 0$.

Quick check · multiple choice

What is the domain of $g(x) = 1/(x-2)$?
A All real x
B $x \neq 2$
C $x > 2$
D $x \geq 2$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Consider $f(x) = \sqrt{5-2x}$. [6]
- (a) State the largest possible domain of f .
 - (b) State the range of f over this domain.
 - (c) Consider $h(x) = (x+1)/(x-4)$. State the domain of h and the equation of any horizontal asymptote (relevant to its range).

Final answer: _____



Answers and explanation

Question 1 [6]

- (a) Need $5-2x \geq 0 \rightarrow x \leq 5/2$.
(b) Since $\sqrt{(\dots)} \geq 0$, range is $f(x) \geq 0$.
(c) Domain: $x \neq 4$. As $x \rightarrow \pm\infty$, $h(x) \rightarrow 1$, so horizontal asymptote $y=1$; range is $h(x) \neq 1$ (all reals except 1, since h never equals 1).

Quick check answer

B. The denominator cannot be zero, so $x \neq 2$ must be excluded; all other reals are valid.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.